

DAY 11 INTERNSHIP REPORT — ADVANCED FILTERING & LOSS ANALYSIS

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Executive Overview

On Day 11 of the internship, the focus was entirely on deep-dive data filtering and root-cause analysis of loss-making segments. Using advanced Excel filtering criteria, targeted analytical queries were constructed to investigate regional profitability issues, evaluate the financial impact of aggressive discounting strategies, and isolate underperforming product sub-categories within the sales dataset.

1. Data Filtering Methodology

To isolate critical insights from the raw transaction data, multi-layered filters were applied combining categorical dimensions (Region, Sub-Category) with numerical thresholds (Profit < 0, Discount >= 40%). This approach allowed for the rapid extraction of subsets specifically tied to revenue leakage.

2. Targeted Analytical Queries & Resolutions

#	Question	Filters Used	Answer Result	Explanation
1	Region with most loss lines?	Profit < 0 + Region	Central (741)	Central Region has a much higher number of losses compared to other regions.
2	Loss \$ at discount >= 40%?	Discount >= 0.4 + Profit < 0	-\$125,346	Applying discounts at or above 40% has resulted in a massive accumulated loss of -\$125,346.
3	Loss-making Tables Lines?	SubCategory=Tables + Profit < 0	203	The count of individual transaction lines showing losses specifically for the Table SubCategory is 203.
4	Top sub-category by sales?	SubCategory + Category	Phones (\$3,30,007)	By total sales volume, Phones rank as the number one sub-category.
5	Deepest discount & outcome?	Discount = 0.8 (+sort)	80%, 0 profitable lines	When discounts are provided at the highest percentage (80%), the store attributes 100% loss for it, yielding zero profitable lines.

3. Business Implications & Actionable Insights

Observation Area	Business Impact & Strategic Takeaway
Discounting Strategy Flaws	The data confirms a direct correlation between heavy discounting and severe margin degradation. Discounts of 80% generate exactly zero profitable transaction lines. Furthermore, aggregate discounts of 40% or more have wiped out \$125,346 in potential profit. The pricing strategy requires immediate restructuring to cap maximum allowable discounts.
Regional Underperformance	The Central Region is significantly underperforming regarding profitability, accumulating 741 individual loss-making transaction lines. This indicates systemic issues in that specific region, possibly tied to localized over-discounting or high fulfillment costs.
Problematic Inventory (Tables)	The 'Tables' sub-category is a major drain on resources, with 203 separate transactions operating at a loss. Despite overall strong performance in 'Phones' (\$3.3L), the losses taken on bulky items like Tables are severely dragging down net margins.